# Graphplay

## The hidden quotient structure of quantum walks

*A formally-verified design framework for continuous-time quantum dynamics — and a pitch for collaborators*

Graphplayers Crew — pitch draft, May 2026

## Abstract

For twenty years, Tamon's group — with Godsil, Severini, Coutinho, Vinet, Bose, Chan, Zhan, Xie, and a generation of co-authors — has been publishing theorems with the same shape: on this graph family, at this time, with this signing, perfect state transfer or uniform mixing or optimal spatial search emerges as if by analytic miracle. Read one such theorem and it looks like number theory. Read fifty and a structure forces itself on you: every miracle factors through an **equitable partition**, the dynamics restricts to a small invariant subspace, and the apparent magic is a finite-dimensional spectral or arithmetic condition on a $r \times r$ matrix where $r$ is independent of the host size. This is folklore. It has never been named, never systematized, and certainly never machine-checked.

**Graphplay names it: the universal coarse-graining of quantum-walk operators.** We formalize it in Lean 4 + Mathlib across seven mathematical settings — finite graphs, weighted/chiral graphs, operator systems and quantum graphs, graphons, categorical filtered colimits, sheaves over parameter manifolds, and an $\infty$ -categorical envelope — and turn it from folklore into a research program. Three lifting theorems, repeated at every tower, are the entire load-bearing skeleton. Dozens of Tamon-clique theorems fall out as immediate corollaries; dozens of **currently open** theorems open up mechanically by transport along the tower. Hardware-aware quantum-walk design — "given this chip, build this primitive" — becomes a small finite optimization over a quotient matrix, with a Lean-checkable certificate of correctness. The Lean development covers Towers 1–5 and has scaffolded statements at Towers 6 and 7; roughly 120 sorries remain, most of them in dowsing-rod files whose **statements** are precise and load-bearing. We want collaborators.

## 1. What is actually new

The thing to notice is that **the structure was always there.** It is not a clever encoding we are imposing on the literature. Bachman–Tamon (arXiv:1108.0339) write PST on quotient graphs. Ide–Narimatsu (arXiv:2209.07688) write equitable-partition spatial search. Chan–Coutinho–Tamon–Vinet–Zhan (arXiv:1907.04729) write fractional revival on association schemes. Levine–Mesapam–Mustico–Tamon–Tucker–Zhan (arXiv:2605.04414) write chiral mixing speedup on signed complete graphs. Xie–Tamon (arXiv:2301.07251) write infinite-tail search optimality. Read with fresh eyes:

> **All five papers are computing the same three quantities on a small finite Hermitian matrix — the quotient by an equitable partition — and then lifting a cell-uniform conclusion to the host.**

The host varies: simple graphs, weighted complex Hermitian graphs, signed graphs, association schemes, graphs with infinite tails. The primitive varies: PST, mixing, search, fractional revival, transfer. The lift varies, slightly, in how it handles marked vertices or boundary cells. But the **shape of the argument** is invariant. Once you see it, you cannot unsee it.

*The headline insight, in one sentence.* Continuous-time quantum-walk operators with engineered dynamical primitives are controlled by a hidden quotient structure that does not care about the ambient category: pick any of {finite graph, weighted/chiral graph, operator system, graphon, filtered colimit, sheaf, $\infty$-category}, write the equitable partition there, and the same three lifting theorems hold.

We call this the **universal coarse-graining** of quantum-walk operators. The three lifting theorems, written once, are:

1. **Spectral lift.** $\operatorname{spec}(A/\pi) \subseteq \operatorname{spec}(A)$, with cell-constant eigenvector lift via the characteristic isometry $P$.

2. **Dynamical restriction.** The unitary $\exp(-itA)$ commutes with the cell-uniform projector $\Pi := P(P^*P)^{-1}P^*$, and acts on range($\Pi$) as $\exp(-it(A/\pi))$.
3. **Primitive transport.** For any cell-uniform initial and target data, PST / mixing / search / fractional revival on $(A, \pi)$ holds iff it holds on the small finite matrix $A/\pi$ (with a one-line refinement for marked vertices).

These are **the** three theorems of Tower 2. Towers 3–7 are the same three theorems with different analysis. That is the entire architecture.

*Why this is new.* Each tower-2 theorem is folklore at the finite-graph level (Godsil, Bachman–Tamon, Ide–Narimatsu). The **statement that they are all the same theorem** — the polymorphism over the host category — has never been written down. The **categorical filtered-colimit preservation** (Tower 5) that makes "quasi-infinite" quantum walks tractable is new. The **graphon-level** lift (Tower 4) that extends Bick–Sclosa (arXiv:2110.13686) from real dynamics to unitary CTQW is new. The **operator-system / quantum-graph** lift (Tower 3) that puts Bose–Mesner association schemes and Duan–Severini–Winter quantum graphs into the same skeleton is new. The mechanization in Lean 4 + Mathlib — with named theorems, machine-checked proofs (where filled), and machine-readable holes (where not) — is new.

## 2. Two illustrative theorems, stated precisely

We give two examples to make the universal coarse-graining concrete.

*Theorem (Bachman–Tamon arXiv:1108.0339, as a Tower-2 corollary).* Let $G$ be a finite simple graph and $\pi$ an equitable partition with characteristic matrix $P$. If $u, v \in V(G)$ are such that $\{u\}$ and $\{v\}$ are cells of the refinement $\pi_{uv}$, and PST occurs between $\{u\}$ and $\{v\}$ in the quotient $A_{\pi_{uv}}$ at time $\tau$, then PST occurs between $u$ and $v$ in $G$ at time $\tau$.

This was published in 2011 as a structural result with a several-page proof. In Graphplay it is `EquitablePartition.pst_lift` applied to the marked-vertex refinement, and is a three-line argument from clauses (ii) and (iii) of the Tower-2 single load-bearing theorem.

*Theorem (Xie–Tamon arXiv:2301.07251, as a Tower-2 + Tower-5 corollary).* Let $G_n = K_n + P_n$ be the family of complete graphs with a path tail of length $n$. The partition $\pi_n = \{K_n, \text{path-end}, \text{path-interior}\}$ is equitable. The quotient matrix $A_{\pi_n}/\pi_n$ is a $3 \times 3$ Hermitian matrix whose entries grow controllably in $n$; its spectral search optimum has a limit as $n \to \infty$ identified as the graphon limit `K + infinite-tail`. No spatial-search algorithm on $G_n$ beats the quotient optimum, for any $n$, and the optimum is realized at $n = \infty$ by the limit graphon.

Again: published as a deep analytic result. In Graphplay it is the filtered colimit of the quotient diagram, plus the cell-uniform restriction of $\exp(-itH_\gamma)$ in $L^2[0,1]$. The infinite-tail optimality is **forced** by the categorical Tower-5 statement `Quotient.preservesFilteredColimits`.

Both proofs sit in `Graphplay/PST.lean`, `Graphplay/Search.lean`, `Graphplay/Categorical.lean`. Several are still `sorry`; the statements are not.

## 3. The seven-tower stack

```
Tower 1  SimpleGraph                     proven, Lean+Mathlib
Tower 2  WeightedGraph (Hermitian/C)     proven, Lean+Mathlib
Tower 3  Operator system / quantum gr.   proven (operator-system
                                         upgrade landed); some sorries
Tower 4  Graphon                         statements complete; ~30% proofs
Tower 5  Categorical (filtered colim.)   statements complete; key theorem
                                         sorried; conceptually clear
Tower 6  Sheaf over parameter manifold   scaffold; awaits Mathlib sheaf-
                                         of-C*-algebras infrastructure
Tower 7  Bicategorical / oo-categorical  scaffold; awaits Mathlib oo-cat
                                         and derived-category libraries
```

Each tower adds analytic content (finite linear algebra; bounded $L^2$ kernel operators; operator-system bimodules; cut-norm graphon limits; filtered colimits of cocones; sheaves of $*$-algebras over a topological base;

coherent idempotents in a homotopy bicategory) but preserves the universal coarse-graining: the equitable partition lifts, the cell-uniform subspace is invariant, the quotient is small.

The **operator-system upgrade** (Tower 3, just landed) is worth flagging separately. Quantum graphs in the Duan–Severini–Winter sense (arXiv:1002.2514) are self-adjoint bimodules over a finite-dimensional $C^*$-algebra; the spine extends to them by treating the equitable partition as a coarsening of the bimodule's diagonal subalgebra. We hope to upstream the cleanest pieces of this — in particular the non-commutative coherent-algebra machinery (`Graphplay/Dowsing/NonCommutativeCoherent.lean`) — to Mathlib. **If you want to contribute Mathlib infrastructure for operator systems and quantum graphs, this is a clean entry point.**

Towers 6–7 are honestly blocked. Tower 6 needs sheaves of unital $*$-algebras over a topological base in a Mathlib-compatible form; Tower 7 needs the $\infty$-category and derived-category libraries that are themselves under active development. We have written the statements down in `Tower6.lean` and `Tower7.lean` so that the proofs become tractable as the surrounding library matures. **We expect Tower 6 to unblock within 12–18 months;** Tower 7 is a multi-year wait.

## 4. Graphplay as a model-based cognitive architecture

A useful sanity check on whether the framework is "real": map it against the universal pattern by which minds build models of reality.

Cognition — across philosophy of science, cognitive science, and modern theory of computation — runs roughly as Observation → Representation → Abstraction → Decomposition → Relational modeling → Inference → Simulation → Evaluation → Optimization, guided by universal principles (Compression, Invariance, Conservation, Symmetry, Compositionality, Emergence) and grounded in formal foundations (logic, information theory, probability, computation, dynamical systems, category theory).

Graphplay maps to this picture without strain:

| **Cognitive stage / principle** | **Graphplay realization** |
|---|---|
| Representation | Choose tower; build `WeightedGraph V` |
| Abstraction | Find equitable partition `EquitablePartition pi` |
| Decomposition | Quotient + orthogonal-fiber decomposition |
| Relational modeling | `GraphBundle Q H B` / k-ary CSP / template |
| Inference | Spectral lift `spec(A/pi) subset spec(A)` |
| Simulation | CTQW evolution `exp(-i t A)` |
| Evaluation | Lean certificate / Mathlib proof |
| Optimization | Chiral-signing optimization on the quotient |
| Compression | Equitable partition *is* lossy compression |
| Invariance | Cell-uniform subspace is `A`-invariant |
| Conservation | Projector $\Pi$ commutes with $A$ (Noether-style) |
| Symmetry | Phantom symmetry / equitable partition |
| Compositionality | Bundles compose (`Tot` is a functor) |
| Emergence | Quotient dynamics *is* emergent dynamics |

We do not claim that quantum-walk design **is** cognition. We claim something weaker and more useful: **the universal coarse-graining of quantum-walk operators is doing exactly what every working scientific model does — compress a high-dimensional dynamical object via an invariant, conservative, symmetric, compositional decomposition into a small one — and the equitable partition is the load-bearing instance of that move for quantum walks.** This is why the framework feels inevitable, and why the same architecture keeps reappearing at each tower.

## 5. Why this matters in practice

Three things become tractable that were not.

*(a) Hardware-aware quantum-walk design.* Take a real chip — IBM heavy-hex (Eagle, Heron, Condor), or Microsoft's Majorana-1 topological qubit, or a Rydberg-atom array — and ask: what quantum-walk primitives can it natively run? In the universal coarse-graining language, this is:

1. Disassemble the chip Hamiltonian into a `GraphBundle Q H B`.
2. Identify the equitable partition $\pi$ forced by the hardware's symmetry.
3. Read the quotient $A/\pi$ as a small Hermitian matrix — typically $2 \times 2$, $3 \times 3$, or $4 \times 4$.
4. Verify in Lean that the primitive of interest (PST, mixing, search) emerges on the quotient and lifts.

The companion notes `applied_ibm_heavy_hex.md` and `applied_majorana1.md` do exactly this. Heavy-hex has a 2-cell data/flag equitable partition with quotient $\begin{pmatrix} 0 & 3 \\ 2 & 0 \end{pmatrix}$ of spectral radius $\sqrt{6}$; cell-uniform PST happens at time $\pi/\sqrt{6}$ using only the chip's native couplings. Majorana-1 has a parity-sector equitable partition whose cells are joint-parity sectors of the tetron Clifford algebra; **topological protection** (flatness of the U(1) connection on the parameter sheaf) becomes exactly the equitable-partition preservation condition. Both disassemblies are Lean-formal and machine-checkable; both surface falsifiable predictions about chip behavior under noise.

*(b) The Tamon-corpus theorems unify.* Roughly two dozen published theorems become corollaries of the three lifting theorems applied at the right tower. The bundle engine `Graphplay/Bundle.lean` realizes Ge–Greenberg–Perez–Tamon (arXiv:1009.1340) products as corners. The graphon limit theorem (`Graphon/Limit.lean`) realizes Xie–Tamon (arXiv:2301.07251) and the Bernard–Tamon–Vinet–Xie (arXiv:2211.14704) tails-transfer companion as filtered-colimit instances. The chiral lift (`Chiral.lean`) makes the Levine et al. (arXiv:2605.04414) mixing speedup into a small finite optimization on the signed quotient. Fractional revival on association schemes (Chan–Coutinho–Tamon–Vinet–Zhan, arXiv:1907.04729) is the commutative case of the operator-system lift. The Ide–Narimatsu (arXiv:2209.07688) search refinement is the marked-vertex refinement of the Tower-2 single load-bearing theorem.

*(c) Open theorems extend mechanically.* Once a theorem is on the spine, you get its tower-$N+1$ analogue **for free** in the sense that the statement transports along the tower, and the proof is "the same proof under the new analysis." The eighteen dowsing-rod files in `Graphplay/Dowsing/` and `Graphplay/Integrations/` each pick out one such open extension. Examples:

- **D1 ChiralBundlePST.** Chiral bundle PST: the chiral analogue of Bachman–Tamon, with the Levine et al. signing-speedup as a corollary.
- **D3 ChiralGraphon.** Graphon-level chiral mixing: the Tower-4 quasi-infinite limit of the Levine et al. speedup, plus a gauge-theoretic reading of cross-constant signings as flat U(1) connections.
- **D7 HypergraphPST.** Hypergraph CTQW; $k$-ary relational extension of the Tower-2 spine.
- **D9 Conjecture93** (flagship falsifiable). A consistent sequence $(G_n, \pi_n)$ admits **both** a graphon quasi-infinite limit **and** a strict chiral mixing speedup, iff the limit Bose–Mesner algebra coincides with the limit partition-projector algebra. **Status:** weak version expected true; strong version expected false; three confirmed both-sides-false witnesses; one candidate strong-counterexample.

The eight integration files extend outward: TQFT (modular tensor categories and Heawood envelopes), RMT (random graphons with deterministic equitable spectra), MERA / PEPS tensor networks (coarse-graining layers as equitable cell maps), optimal transport on graphons, mean-field games on graphon hosts, combinatorial Hodge theory on the relational tower, lattice gauge theory through chiral signings, and Weisfeiler–Leman refinement as equitable saturation. Each is a research-program seed, not a finished theorem.

## 6. The research program

What the framework affords, scaffolded:

*Known and proved (Lean-checked, modulo flagged sorries):* the three Tower-2 lifting theorems, the bundle equitability criterion (regular fibers + biregular couplings), the bundle-PST corollary chain recovering Ge–Greenberg–Perez–Tamon, the chiral characteristic-isometry intertwining, the search marked-vertex refinement, the equitable-partition characterization of commutative coherent subalgebras (folklore + WL refinement chain), and the graphon spectral lift via Szegedy's regularity.

*Known and stated, sorried (Lean-stated, proofs deferred):* the graphon CTQW lift; the graphon search and PST optimality at the quasi-infinite limit; the categorical filtered-colimit preservation of `Quotient`; the operator-system / quantum-graph lift; chiral bundle PST; the bundle-level uniform-iff PST; non-commutative coherent algebras as fixed points of quantum WL refinement; the operator-system quantum-chromatic-number bridge.

*Conjectural with worked examples (statements precise; proofs and falsifiers open):* Conjecture 9.3, the Bose–Mesner ≡ chiral ≡ graphon iff; chiral honeycomb signings on heavy-hex generalising Levine et al.'s $K_4$ unitary signing; chiral PGST on Heawood bundles; quantum graphon limits; the Anantharaman–Sabri-style chiral connection in Benjamini–Schramm limits.

*Sheaf / oo-categorical (Tower 6–7, scaffold only):* quasiparticle-poisoning as a $H^1$-class in the parameter sheaf; functoriality of gate design across chip generations as filtered colimits of chip-extension arrows; bicategorical truncation for measurement-based braiding with Pauli-correction 2-cells.

There are roughly **seven open Conjecture-9.3-style flagship questions** on the program, each tractable for a strong PhD student, each with worked examples backing the claim it is the right question.

# 7. Honest assessment

The Lean development is 45 files, 98% of which build green. There are approximately 120 high-value `sorry` holes — the rest are scaffolding sorries in dowsing-rod files where the **statement** is the contribution, not the proof. Tower 1–2 are largely proven. Tower 3 has the operator-system upgrade and several genuine theorems, alongside the non-commutative coherent-algebra sorries. Tower 4 has clean statements and 30% complete proofs of the graphon CTQW lift. Tower 5 has the filtered-colimit statement and a key theorem sorried. Towers 6–7 are honest scaffolds, awaiting Mathlib library development.

We need:

1. **Mathematicians** who care about quantum walks, operator algebras, graphons, or coherent algebras, who can take a dowsing-rod statement and produce a real proof. The Tower-2 lifts are mostly classical results awaiting Mathlib mechanization; the Tower-3/4/5 lifts are genuinely open in the form we state.
2. **Physicists / quantum-hardware people** who can validate the engineering claims. The Majorana-1 and heavy-hex disassemblies make falsifiable predictions about chip behavior; we want adversarial reading.
3. **Mathlib contributors** who can help upstream the operator-system, quantum-graph, and graphon-spectrum infrastructure. The cleanest pieces belong in Mathlib, not in Graphplay.
4. **Theory of computation / category theorists** who can sharpen the Tower-5 filtered-colimit statement and prepare the Tower-7 $\infty$-categorical envelope.
5. **Anyone with a hardware platform.** Pick your favorite quantum-walk hardware — Rydberg arrays, photonic networks, trapped ions, superconducting fabrics — and disassemble it. The disassembly is a paper.

The project is **exploratory**. Not every dowsing-rod theorem will survive contact with proof; some will be sharpened, some will turn out false, some will yield a stronger statement nobody anticipated. The machine-readable certificate manifest emitted by `Graphplay.Toolkit.Report` makes the proven / unproven split transparent at all times. We welcome collaborators at any level: from "I have a chip" to "I have a sharp eye for spectral analysis" to "I want to learn Lean by proving a theorem nobody has proved."

# 8. How to get involved

The repository: `github.com/<owner>/graphplay` (cloneable; `lake build` on a current Mathlib).

*A 30-minute reading plan, by audience:*

- **Tamon postdoc / CTQW mathematician.** `paper/quasi_infinite_adjoint_v3.typ` §§2–7. Then pick a dowsing-rod file in `Graphplay/Dowsing/`.
- **Mathlib contributor.** `Graphplay/QuantumGraph.lean`, `Graphplay/Dowsing/NonCommutativeCoherent.lean`, and any operator-system statement. These are upstream candidates.

- **Categorical / homotopy theorist.** Graphplay/Categorical.lean and Graphplay/Tower7.lean.
- **Graphon / graph-limits theorist.** Graphplay/Graphon/Limit.lean and paper/quasi_infinite_adjoint_v3.typ §5.
- **Quantum-hardware engineer / physicist.** paper/applied_majorana1.md and paper/applied_ibm_heavy_hex.md.
- **Lean-curious newcomer.** Graphplay/StdLib/ — the standard-library entries (Cayley, complete-multipartite, Hamming, hypercube, path) are small, well-shaped, and several have open sorrys on classical spectral identities.

The actionable surface is: pick a dowsing-rod statement, or pick a hardware platform, or pick a stdlib entry. Each is self-contained. Each gets you a paper, or a Mathlib contribution, or a worked example we will gladly fold into the spine.

## 9. One paragraph for the timeline

The spine has two decades of literature underneath it; we are naming a structure that has been hiding in plain sight since Godsil's original equitable-partition spectral lemma. We are not the first to notice that quotients matter; we are the first — as far as we can tell — to write down **the** lift theorem polymorphically, mechanize it, push it across seven mathematical settings, and treat the catalog of Tamon-corpus theorems as a unified body to be derived rather than collected. If the framing is right — and the cognitive-architecture sanity check, and the unification of two dozen disparate analytic miracles, and the operator-system upgrade landing cleanly, all suggest it is — then the next two to five years should see a steady stream of formerly-hard theorems falling out of the spine, and quantum-walk hardware design becoming the small finite optimization it always secretly was.

---

*Repository:* github.com/<owner>/graphplay. *Main paper:* paper/quasi_infinite_adjoint_v3.typ. *Research program:* paper/research_program.typ. *Applied disassemblies:* paper/applied_majorana1.md, paper/applied_ibm_heavy_hex.md. *Contact:* DM the project lead, or open an issue on the repo. Collaborators welcome at any level.